

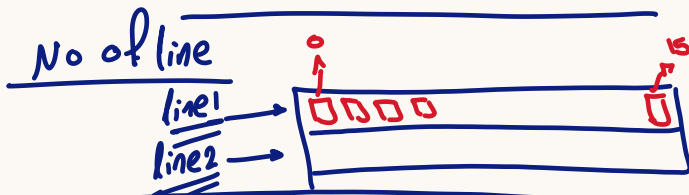
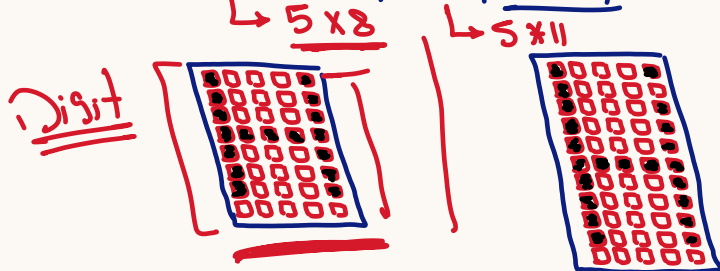
Display H.W

- ① LED → Notify
- ② Segment → Display number.
- ③ Screen → Messages

↳ LCD :: Liquid crystal Display module

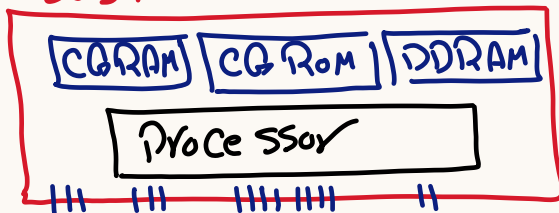
LCD → ① Size :: 16×2 | 20×4 | 16×4
 Digits
 number line

↳ Digits → write 1 character.
 No of Pixel

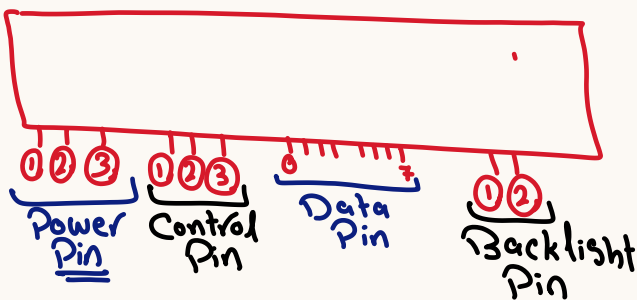


LCD :: ① Pinout ② over view Module ③ work Flow

LCD Module



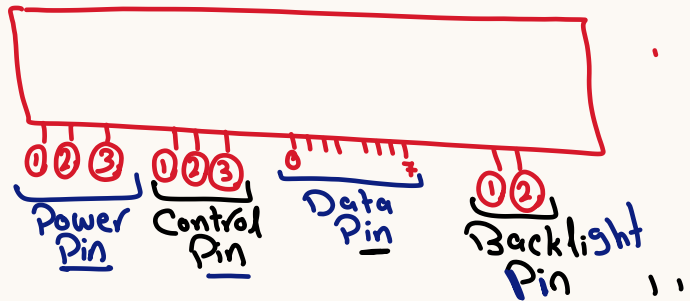
* Pinout



① Power Pin

- ↳ ① Vss | gnd → gnd
- ↳ ② VDD | Vcc → 5V
- ↳ ③ Vo → D Pot
 Contrast

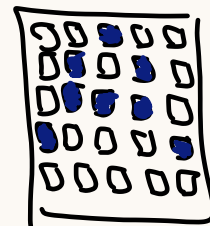
* Pinout



① Power Pin

- ↳ ① Vss | gnd → gnd
- ↳ ② VDD | Vcc → 5V
- ↳ ③ Vo → D Pot
 Contrast → Darkness

* Note :: LCD → all Pixel is turn on
 if you need write turn off
 for Pixel →



② Control Pin

MC → RS :: Register select write lcd
 Data for Display RS=1
 Data as Instruction RS=0

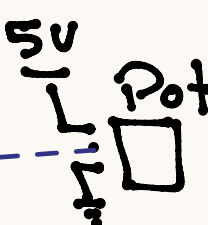
MC → R/W → Read | write
 (1) (0)

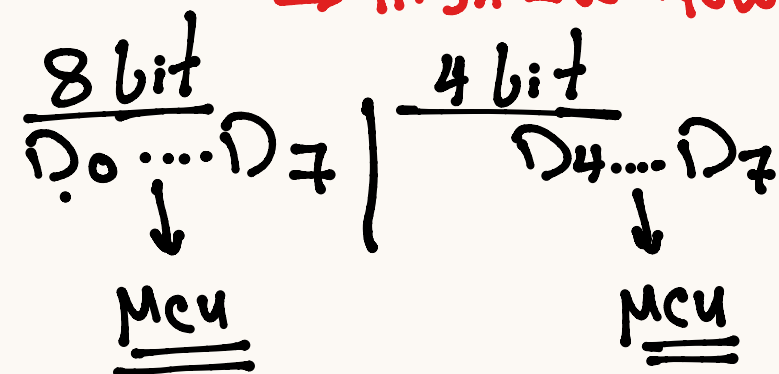
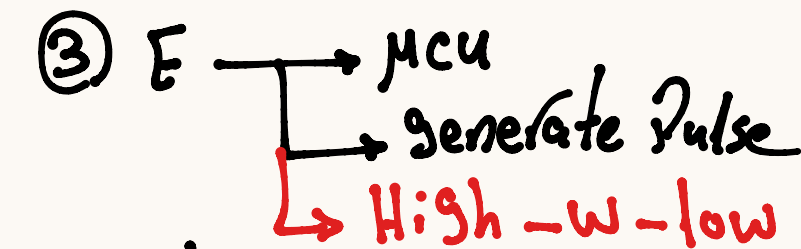
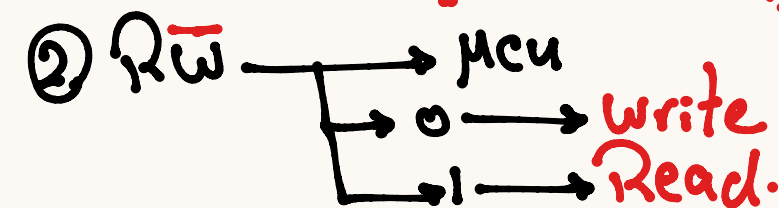
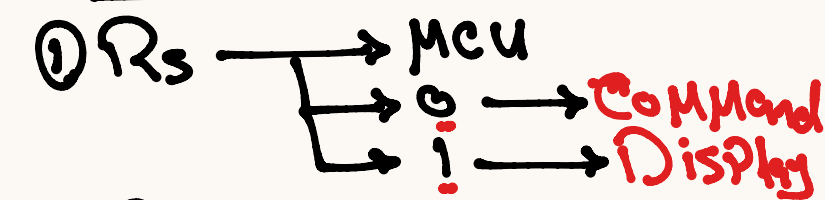
MCU → Enable :: High wait Low
 generate clock for lcd processor
 clk

③ Data Pin (1Byte)

- ↳ Data for display or Instruction
- ↳ Can be 8 bit (8 wire) with MCU
- ↳ Can be 4 bit (4 wire) with MCU

Summary Pinout

- ① gnd
- ② Vcc → 5v
- ③ Vo → 



① B1A → 5v

② B1K → gnd

Overview Module

① DDRAM

Data Display Ram

* LCD → Display character only

* Max Data → 1 Byte (ASCII)

④ Back light

* Power for screen

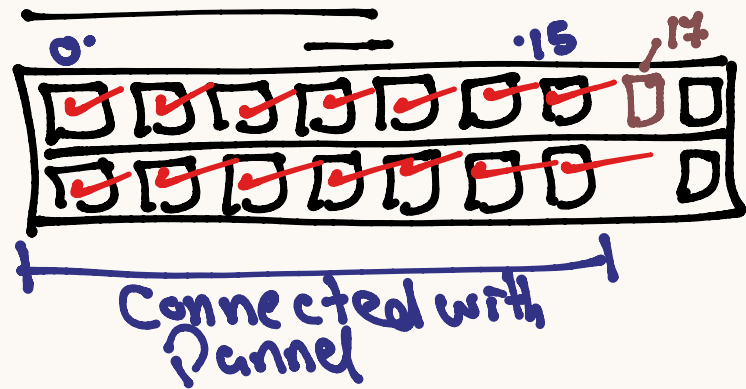
* ① B1A → Back light Anode
* ② B1K → Back light Cathode



LCD('A');

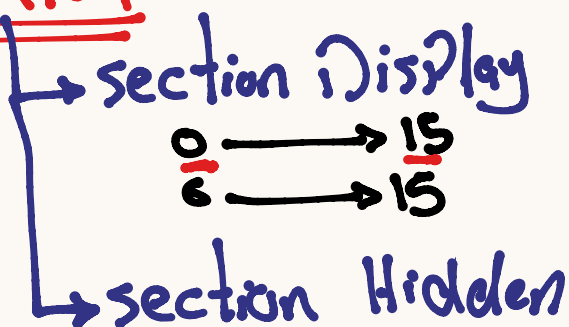
3

* DDRAM

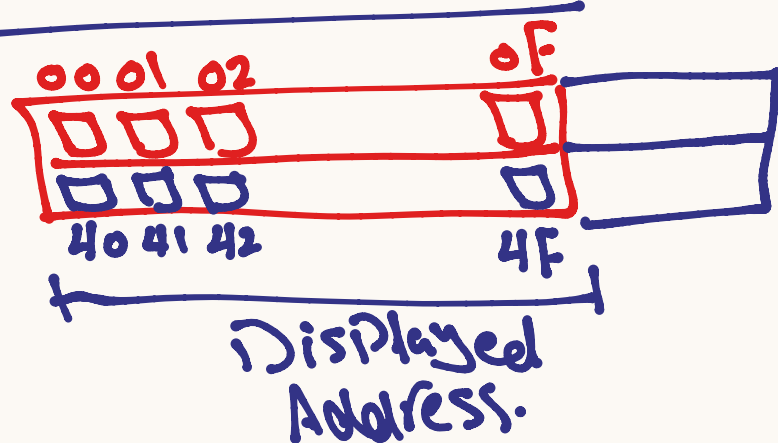


* Connect with led Panel

* DDRAM → have two Section



* DDRAM Address



② CGRAM

Character generator
Rom

* store the Display way
for all ASCII

* Address → with ASCII
No

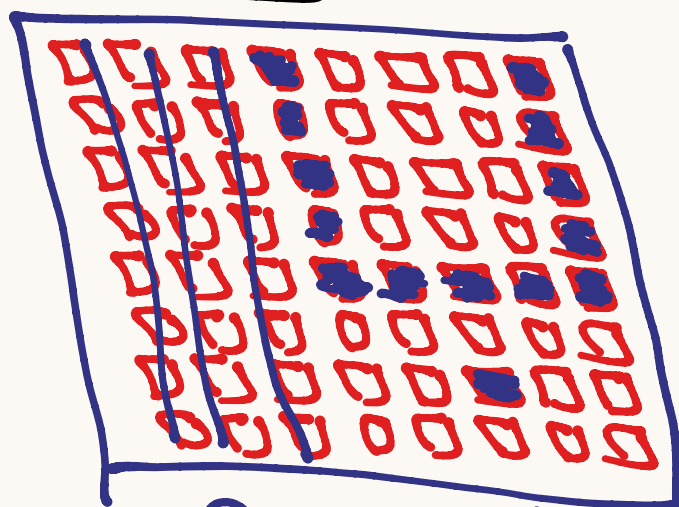
* CGRAM

Character generator
Ram

→ Memory to store special
Character

→ Must be send Display
way.

→ CGRAM → 8 location
→ each location size
8 byte.
→ each location
Can store 1 char
only.



→ CGRAM Address

* location 0 → 8 byte → 0x00-0x07

* location 1 → 0x08

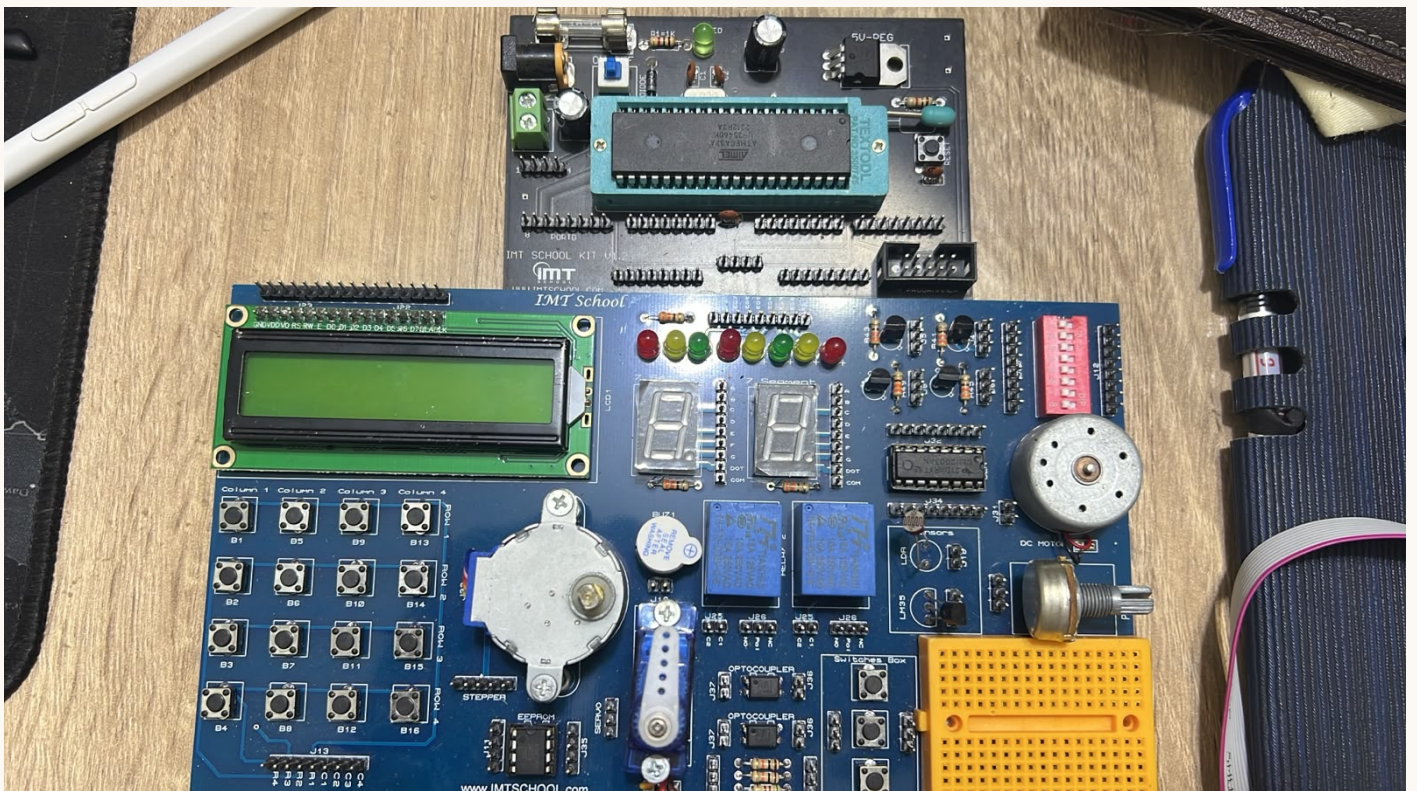
* Processor of lcd

if → Display character on lcd

Processor → Data from Data Pin
CGRAM ← ASCII

Take Display way
for this ASCII

→ DDRAM
store Display
way



Function

void LCD_sendCommand(48 c)

4
Rs → 0
Rw → 0
data → c
E → 1
wait (1msec)
E → 0

5

void LCD_writecharacter(48 data)

4
Rs → 1
Rw → 0
dataPin → data
E → 1
wait (1msec)
E → 0

5

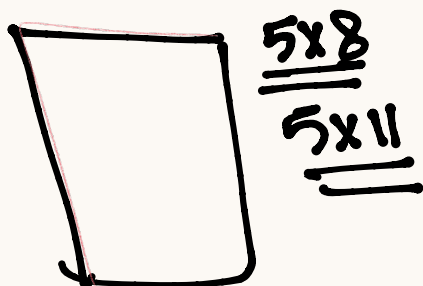
Init sequence

8 bit Mode Init

- ① Power on
- ② wait 30msec.
- ③ Command Function set.
- ④ wait 39μsec.
- ⑤ Command Display on OFF
- ⑥ wait 39μsec.
- ⑦ Command Display clear
- ⑧ wait 1.53msec
- ⑨ Command Entry Mode
- ⑩ End of Initialization.

void LCD_Init(

1



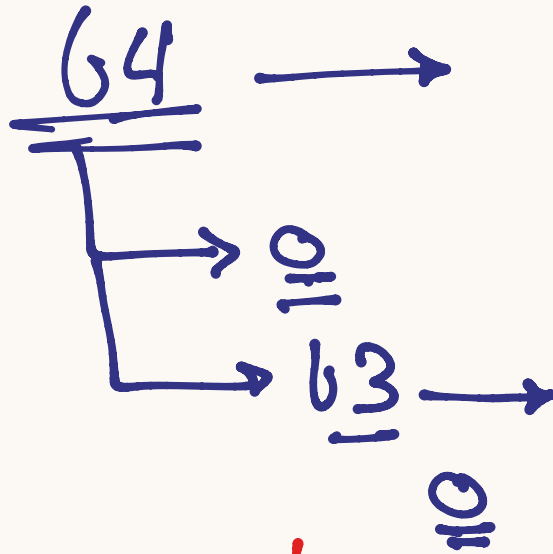
LCD Display on 16

- ① D=0 C x B x
- ② D=1 C=0 B x
- ③ D=1 C=1 B=0
D=1 C=1 B=1

* Set CGRAM Address

↳ Processor → Points on specific
in CGRAM Address

↳ Max Address in CGRAM



6 bit
11111

Address 10

For ex: → location → CGRAM → 3
Address ⇒ location No * 8
 $3 * 8 = 24 = \boxed{0x18}$

Command ⇒

01011000 →

Address 0601000000

0x18 | 0x40

01011000.



lcd Panel